

PRIMITIVE-RECURRENCE AVOIDANCE SHIFTS: SINGER-CYCLE COMPRESSION AND A DELAYED PERIODIC ANOMALY

ANONYMOUS

ABSTRACT. Fix a primitive degree- r recurrence over $\mathbb{F}_q$ and forbid its discrepancy from vanishing at every time. For $q \geq 3$ and $r \geq 2$, the resulting $(q-1)$ -regular shift of finite type is mixing and has a unique, uniform measure of maximal entropy. Put

$$L = q^r - 1, \quad H = q^{r-1} - 1, \quad D = (q-1)^H.$$

Although its state matrix has size q^r , finite Fourier transform reduces it to one scalar block and one weighted Singer cycle. We obtain

$$\det(tI - A) = (t - (q-1))(t^L - D), \quad \zeta(z) = \frac{1}{(1 - (q-1)z)(1 - Dz^L)}.$$

Consequently every fixed-point count before time L is exactly $(q-1)^n$, the count of the full $(q-1)$ -shift; the first discrepancy occurs at L and has size LD . Thus F_1 and the location of the first anomaly recover (q, r) . The same algebraic formulas remain valid at $q = 2$, where mixing fails and the system becomes one fixed orbit plus one orbit of length $2^r - 1$. Exact controls build the full adjacency matrices over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4$, and $\mathbb{F}_5$ and verify the Fourier-cycle ingredients, full characteristic polynomial, periodic counts, and recovery statement.

1. INTRODUCTION

A primitive linear feedback shift register has the simplest possible nonzero state dynamics: its companion matrix cycles through all nonzero states. That deterministic cycle structure and its relation to primitive polynomials and Singer cycles are classical and remain active subjects in coding theory [2, 3]. We change the local rule in one specific way. Instead of enforcing zero recurrence error, we forbid zero error and allow every nonzero error at every step. The deterministic register then becomes a regular shift of finite type with $q-1$ choices from each state.

The change is small at the level of the defining recurrence but not at the level of periodic data. A direct determinant has degree q^r . The useful coordinates are additive characters: the zero character gives the entropy root, while a primitive companion matrix puts every nonzero character into one orbit. Along that orbit the Fourier weights are $q-1$ on one coordinate hyperplane and -1 off it. Counting the hyperplane intersections produces the complete determinant with two factors.

The contribution is the following scope-delimited theorem package.

- We give an explicit finite affine skew-product model and a full Fourier/block proof of the two-factor characteristic polynomial.
- We compute every fixed-point count, every least-period orbit count, and the Artin–Mazur zeta function. Periodic data agree with the full $(q-1)$ -shift until the sharp time $q^r - 1$.
- We prove mixing and identify the unique maximal-entropy measure for $q \geq 3$, isolate the reducible binary boundary, and recover (q, r) from the first periodic anomaly.

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The ownership boundary is important. We do not claim primitive polynomials, Singer cycles, or the cycle structure of deterministic linear feedback registers [2, 3]. We also do not claim the determinant formula for a general shift of finite type [1] or the Parry maximal-entropy construction [4]. The residual object is the nonzero-error relation and the weighted Fourier cycle it creates. A bounded search found no direct source for this combined formula and delayed-anomaly theorem, but no absolute novelty or priority claim is made. External posting and submission remain on hold pending specialist literature and priority review.

2. THE AVOIDANCE SHIFT AND ITS STATE MODEL

Let q be a prime power and $r \geq 2$. Fix a primitive polynomial

$$(1) \quad f(t) = t^r - c_{r-1}t^{r-1} - \cdots - c_1t - c_0 \in \mathbb{F}_q[t]$$

and its companion matrix

$$(2) \quad C = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ c_0 & c_1 & c_2 & \cdots & c_{r-1} \end{pmatrix}.$$

Primitivity means that a root of f generates $\mathbb{F}_q^\times$. In particular, C has order

$$L = q^r - 1$$

and acts transitively on $\mathbb{F}_q^r \setminus \{0\}$.

Definition 2.1 (Primitive-recurrence avoidance shift). The shift $X_{q,r,C} \subset \mathbb{F}_q^{\mathbb{Z}}$ consists of the sequences x for which

$$(3) \quad e_n(x) := x_{n+r} - \sum_{j=0}^{r-1} c_j x_{n+j} \in \mathbb{F}_q^\times \quad (n \in \mathbb{Z}).$$

It is equipped with the left shift σ .

For $s = (s_0, \dots, s_{r-1})^\top \in \mathbb{F}_q^r$, write $\mathbf{e}_r = (0, \dots, 0, 1)^\top$. The r -block presentation has state adjacency matrix A , indexed by $\mathbb{F}_q^r$, with

$$(4) \quad A_{s,t} = 1 \iff t = Cs + u\mathbf{e}_r \text{ for some } u \in \mathbb{F}_q^\times.$$

The overlap coordinates in (2) show that this presentation is conjugate to $X_{q,r,C}$, not merely a cover.

There is also a useful global description. Let $E = (\mathbb{F}_q^\times)^\mathbb{Z}$ and define

$$(5) \quad T(s, e) = (Cs + e_0\mathbf{e}_r, \sigma e) \quad \text{on } \mathbb{F}_q^r \times E.$$

Proposition 2.2 (Finite affine skew product). *The map*

$$x \mapsto ((x_0, \dots, x_{r-1})^\top, (e_n(x))_{n \in \mathbb{Z}})$$

is a topological conjugacy from $(X_{q,r,C}, \sigma)$ to $(\mathbb{F}_q^r \times E, T)$. Moreover, every row and every column of A has sum $q - 1$.

Proof. Given a state and a two-sided error sequence, (3) determines the future recursively. Since $c_0 \neq 0$, or equivalently C is invertible, the same equation determines the past. This construction is inverse to the displayed map and intertwines the dynamics with (5).

For a fixed source state, the $q - 1$ nonzero values of u in (4) give distinct targets. For a fixed target t and each $u \neq 0$, invertibility gives the unique source $C^{-1}(t - u\mathbf{e}_r)$. Thus both degrees equal $q - 1$. $\square$

3. FOURIER COMPRESSION

Put

$$(6) \quad H = q^{r-1} - 1, \quad D = (q - 1)^H.$$

The exponent H counts the nonzero points of a coordinate hyperplane.

Fix a nontrivial additive character $\psi: \mathbb{F}_q \rightarrow \mathbb{C}^\times$. For $\xi, s \in \mathbb{F}_q^r$, set

$$\chi_\xi(s) = \psi(\langle \xi, s \rangle).$$

The q^r functions χ_ξ form a basis of the complex functions on $\mathbb{F}_q^r$. Indeed, if $\delta \neq 0$, choose a nonzero coordinate δ_j , choose $a \in \mathbb{F}_q$ with $\psi(a) \neq 1$, and put $s_j = a/\delta_j$ with the other coordinates zero; then $\psi(\langle \delta, s \rangle) \neq 1$. Thus the character pairing is nondegenerate even when q is not prime. Regard A as the forward transfer operator

$$(A\varphi)(s) = \sum_{u \in \mathbb{F}_q^\times} \varphi(Cs + u\mathbf{e}_r).$$

Lemma 3.1 (Weighted dual permutation). *For every $\xi \in \mathbb{F}_q^r$,*

$$(7) \quad A\chi_\xi = w(\xi_r)\chi_{C^\top \xi}, \quad w(a) = \begin{cases} q-1, & a = 0, \\ -1, & a \neq 0. \end{cases}$$

Proof. Substitution into the transfer operator gives

$$(A\chi_\xi)(s) = \psi(\langle C^\top \xi, s \rangle) \sum_{u \in \mathbb{F}_q^\times} \psi(u\xi_r).$$

The last sum is $q-1$ when $\xi_r = 0$. When $\xi_r \neq 0$, multiplication by ξ_r permutes $\mathbb{F}_q^\times$, and the sum over the whole field is zero; deleting the term at zero leaves -1 . $\square$

Theorem 3.2 (Two-block characteristic polynomial). *For every prime power $q \geq 2$ and every $r \geq 2$,*

$$(8) \quad \boxed{\det(tI - A) = (t - (q - 1))(t^L - D)}.$$

Equivalently,

$$(9) \quad \det(I - zA) = (1 - (q - 1)z)(1 - Dz^L).$$

Proof. The zero character spans an invariant line, and (7) acts there by $q-1$. Since C is a Singer cycle, $C^\top$ is conjugate to a Singer cycle and therefore has one orbit of length L on the nonzero dual vectors. The remaining Fourier block is a weighted cyclic permutation. Its characteristic polynomial is

$$(10) \quad t^L - \prod_{i=0}^{L-1} w(((C^\top)^i \xi)_r)$$

for any $\xi \neq 0$.

The orbit visits every nonzero vector exactly once. Among those vectors, exactly $q^{r-1} - 1 = H$ have last coordinate zero. The other

$$L - H = q^{r-1}(q - 1)$$

vectors contribute a factor -1 . This number is even: if q is odd then $q-1$ is even, and if q is even then q^{r-1} is even because $r \geq 2$. The product in (10) is therefore $(q-1)^H = D$. Multiplying the zero-frequency factor and the nonzero-frequency factor gives (8). Replacing t by z^{-1} and multiplying by z^{q^r} gives (9). $\square$

The proof identifies the compression mechanism rather than inferring it from traces: one scalar Fourier mode and one weighted cycle account for all q^r eigenvalues.

4. ALL PERIODS, ZETA, AND RECOVERY

Write

$$F_n = \# \text{Fix}(\sigma^n \mid X_{q,r,C}).$$

The block presentation is a conjugacy, so $F_n = \text{tr}(A^n)$.

Theorem 4.1 (Complete periodic data). *For every $n \geq 1$,*

$$(11) \quad F_n = (q-1)^n + \mathbf{1}_{\{L \mid n\}} L (q-1)^{Hn/L}.$$

The Artin–Mazur zeta function is

$$(12) \quad \zeta_{q,r,C}(z) = \exp \left(\sum_{n \geq 1} \frac{F_n}{n} z^n \right) = \frac{1}{(1 - (q-1)z)(1 - Dz^L)}.$$

The number O_n of shift orbits of least period n is

$$(13) \quad O_n = \frac{1}{n} \sum_{d \mid n} \mu(n/d) \left((q-1)^d + \mathbf{1}_{\{L \mid d\}} L (q-1)^{Hd/L} \right),$$

where μ is the Möbius function.

Proof. The L roots of $t^L - D$ have n th-power sum zero unless $L \mid n$. When $n = kL$, that sum is LD^k . Adding the n th power of the scalar eigenvalue $q-1$ proves (11). Summing the contribution of the multiples $n = kL$ in the defining logarithm gives

$$\sum_{k \geq 1} \frac{LD^k z^{kL}}{kL} = \sum_{k \geq 1} \frac{(Dz^L)^k}{k} = -\log(1 - Dz^L),$$

while the scalar mode gives $-\log(1 - (q-1)z)$. This proves (12). Finally, $F_n = \sum_{d \mid n} dO_d$; Möbius inversion gives (13). $\square$

The formula has a sharp identification consequence.

Corollary 4.2 (First-anomaly rigidity). *Let $\Delta_n = F_n - F_1^n$. Then*

$$(14) \quad F_1 = q-1, \quad \min\{n \geq 1 : \Delta_n > 0\} = L = q^r - 1, \quad \Delta_L = LD.$$

Hence the fixed-point sequence recovers

$$q = F_1 + 1, \quad r = \log_q(L+1).$$

In particular, two shifts in this family with identical fixed-point data have the same pair (q, r) .

Proof. Because $r \geq 2$, one has $L > 1$, and (11) gives $F_1 = q-1$. The correction term in that formula is positive exactly at multiples of L , proving all three assertions in (14). The displayed recovery formulas follow. $\square$

The agreement before L is exact, not asymptotic: through time $L-1$ the periodic counts are indistinguishable from those of the full shift on $q-1$ symbols.

5. MIXING, MAXIMAL ENTROPY, AND THE BINARY BOUNDARY

Theorem 5.1 (Mixing and uniform maximal entropy). *Suppose $q \geq 3$. Then $X_{q,r,C}$ is topologically mixing,*

$$(15) \quad h_{\text{top}}(X_{q,r,C}) = \log(q-1),$$

and it has a unique measure of maximal entropy. In the state presentation, that measure has stationary distribution q^{-r} on $\mathbb{F}_q^r$ and transition probabilities

$$(16) \quad P_{s,t} = \frac{A_{s,t}}{q-1}.$$

Equivalently, the recurrence errors form an i.i.d. uniform process on $\mathbb{F}_q^\times$, independent of the uniform time-zero state.

Proof. By [proposition 2.2](#), A is both row- and column- $(q-1)$ -regular. Hence $q-1$ is a Perron eigenvalue with constant left and right eigenvectors. The other eigenvalues have modulus

$$D^{1/L} = (q-1)^{H/L} < q-1,$$

because $q-1 > 1$ and $H < L$. Thus the Perron eigenvalue is simple and is the only eigenvalue on its spectral circle.

For completeness, double regularity turns simplicity into irreducibility. If the directed graph were not strongly connected, a sink strongly connected component would have no outgoing edges. Its total internal row sum would already equal $(q-1)$ times its number of vertices. The equal column sums then leave no incoming edges either, so this component and its nonempty complement would each contribute an eigenvalue $q-1$, contradicting simplicity. An irreducible nonnegative matrix of period d has d eigenvalues on its spectral circle. Therefore $d = 1$, and A is primitive. This proves mixing and [\(15\)](#).

The Parry construction for a primitive matrix is unique [\[4\]](#). Substituting the constant Perron vectors gives the uniform stationary law and [\(16\)](#). Since each labeled error determines one successor and its conditional law is uniform independently of the current state, successive errors are i.i.d.; the same factorization shows independence from the stationary initial state. $\square$

The restriction $q \geq 3$ is necessary.

Proposition 5.2 (Binary boundary). *When $q = 2$, the algebraic formulas in [theorems 3.2](#) and [4.1](#) remain valid with $D = 1$, but the shift is not mixing. It is the disjoint union of one fixed orbit and one orbit of length $L = 2^r - 1$. Consequently it has two ergodic maximal-entropy measures and*

$$\zeta_{2,r,C}(z) = \frac{1}{(1-z)(1-z^L)}.$$

All maximal-entropy measures in the binary case are the convex combinations of these two ergodic periodic-orbit measures.

Proof. There is only one allowed error, so the state map is the permutation $s \mapsto Cs + \mathbf{e}_r$. Since 1 is not an eigenvalue of the primitive companion matrix, $I - C$ is invertible and the affine permutation has a unique fixed point s_* . Translation by s_* conjugates it to C . The latter fixes zero and cycles through all L nonzero states, which proves the orbit decomposition and the measure statement. $\square$

6. EXACT CONTROLS AND SCOPE

The deterministic control program constructs the full state matrices rather than substituting the theorem formulas. It also constructs the dual Singer orbit, counts its coordinate-hyperplane intersections, computes $\det(tI - A)$ by the integer Faddeev–LeVerrier algorithm, checks strong connectivity and period, and compares traces through time $L + 1$. The frozen cases are shown in [table 1](#).

TABLE 1. Exact full-matrix controls. Coefficients are ordered $(c_0, \dots, c_{r-1})$ as in [\(1\)](#). The anomaly column is $F_L - (q-1)^L = LD$.

q	r	coefficients	q^r	L	LD	mixing
2	2	(1, 1)	4	3	3	no (boundary)
3	2	(1, 1)	9	8	32	yes
3	3	(2, 0, 1)	27	26	6656	yes
4	2	(2, 1)	16	15	405	yes
5	2	(2, 2)	25	24	6144	yes

The $\mathbb{F}_4$ lane uses the independent model $\mathbb{F}_2[u]/(u^2 + u + 1)$. Across the five lanes, the script records 258 exact assertions. For example, at $(q, r) = (3, 3)$ it verifies

$$F_{26} = 2^{26} + 6656 = 67,115,520,$$

whereas every F_n for $1 \leq n < 26$ equals 2^n .

The theorem does not cover nonprimitive recurrence polynomials or arbitrary allowed error sets. In those regimes the nonzero dual frequencies can split into several orbits, and the weight product in (10) need not reduce to one factor. That orbit splitting is a different problem. Within the primitive, nonzero-error family, the Fourier proof gives the complete spectrum, dynamics, and recovery statement without an unverified genericity assumption.

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